There is no elementary embedding of a larger free system into a smaller one

Silvia Barbina, Enrique Casanovas

arxiv:2411.13723v3; Creative Commons Attribution 4.0 International

Editorial note. The counted statement is the author's environment at source0.tex lines 1089-1090 and its proof environment opens at line 1091; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

Editorial note. Metadata provenance: the source record of the pinned copy and the licence evidence file are both present; the licence is CC BY 4.0 with status supported, and the card's source_version matches the value recorded in that evidence file.

Editorial scope. One original result of the pinned author manuscript, counted once: no elementary embedding of a larger free system into a smaller one (author label April_11_2024_7).

Allowed standard prerequisites (editorial).

- Free systems on finitely many generators. F_n is the free system on n generators, and it embeds into F_m when n is at most m.. They are the two systems compared.
- Elementarity transfers from the inclusion. If one model is an elementary subsystem of another then the inclusion is an elementary embedding.. It turns the failure of an embedding into a contradiction.

- Surjectivity of elementary self-embeddings. Every elementary embedding of F_n into itself is surjective.. It is the statement contradicted at the end.

Proposition 0.1 Let M, N be models of the theory of free STSs. If $M \leq N$, then $M \preccurlyeq N$.

Proposition 0.2 Let F_n be the free STS with $n \geq 3$ generators. Every elementary embedding $f : F_n \to F_n$ is surjective and, therefore, is an automorphism of F_n.

Proposition 0.3 If $3 \leq n < m$, there is no elementary embedding of F_m, the free STS with m generators, into F_n, the free STS of n generators, despite the elementary equivalence of F_m and F_n.

Proof. Assume $n < m$ and $f : F_m \to F_n$ is an elementary embedding with $f(F_m) = F'_m$. There is a copy F''_m of F_m such that $F_n \leq F''_m$, so $F_n \preccurlyeq F''_m$ by Proposition 0.1. Without loss of generality, $F_m = F''_m$. Let g be the inclusion map $F_n \mapsto F_m$ restricted to F'_m. Then $g \circ f : F_m \to F_m$ is an elementary injection with $g \circ f(F_m) \neq F_m$, contradicting Proposition 0.2.

□